

ICT Course Contents

Unit-1: Introduction to Information and Communications Technology (ICT) and Computer System

- Definition of ICT, basic concepts and terminologies
- Data, Information
- Application of ICT, Advantages and Disadvantages of ICT
- Impact of ICT, Career opportunities in ICT education
- Information Processing Cycle
- Classification of Computers
- The Vital Components of the Systems Unit

Unit-2: Computer Hardware and Software

- Categories of Computer Hardware: Input devices, Processing devices, Output devices, Storage devices, Communication devices
- Main Processing Devices: The Processor, Control Unit, and Arithmetic and Logic Unit
- Software Packages, Operating Systems
- Types and uses of Operating Systems
- Types and uses of Application Software: Educational software, Game software, Graphics software, Database software
- Browsers: Internet Explorer, Google Chrome, Mozilla Firefox, Opera
- Proprietary and Open Source Software

Unit-3: Introduction to Word Processing Application

- Word Processing Packages and Their Uses
- Creating a Document Using a Word Processor
- Saving a Document Using the 'Save As' command
- Editing a Word Document Using Editing Tools
- Copy/cut, paste, Undo and redo, find, replace, wrapping
- Inserting a table, inserting symbols and pictures
- Header, footer, page number, drop cap, word art
- Page Design: watermark, page border

Unit-4: Spreadsheet Application

- Layout: Page setup/print
- Spreadsheet Packages, e.g. Excel
- Importance of Spreadsheet application in Data Management
- Related Concepts and Terminologies (e.g., cell(s), rows, columns, worksheet, workbook)
- Features in the Spreadsheet Window
- Tool Bars: formatting bar, standard bar, formula bar

- Types of Data and Their Uses (e.g., number, date, text, currency)
- Creating and Saving a Workbook
- Constructing and Inserting Simple Formulae and Functions
- Formatting Worksheet Using Formatting Tools
- Draw a Graph/chart
- Editing and Printing Worksheet

Unit-5: Presentation Application [প্রেজেন্টেশন অ্যাপ্লিকেশন]

- Presentation Applications Packages
- Devices used for Presenting
- Importance of Presentation Application
- Principles for Designing Presentations
- Terminologies in Presentation Application (e.g., Slide Layout, Slide transitions, Slide show, etc.)
- Identification of Commonly Used Features of Presentation Application Window: Toolbars, Different presentation view modes
- Prepare a Presentation: Adding elements and formatting slides, slide show, slide transition, animation effects
- Selection of Print Option: Entire presentation, specific slides, handouts, notes pages, outline view of slides, and number of copies

Unit-6: Privacy and Security [গোপনীয়তা ও নিরাপত্তা]

- Introduction to Information Security
- Cybercrime, DoS and DDoS attack
- Key Management, Digital Signature and Certifications
- Privacy, Data Security, Vulnerability, Threat and Risk
- Malware, Social Engineering, Hacking
- Plagiarism, Phishing, Software Piracy, Worms and Viruses
- Spam, Adware, Malware, Spyware, and Ransomware
- Ethics in the Digital World

Unit-7: Using the Internet to Communicate and Accessing Information [ইন্টারনেট ব্যবহার করে যোগাযোগ ও তথ্য আহরণ]

- Introduction to Internet, IP Address, MAC Address
- Internet Services, OSI Reference Model, TCP/IP Protocol Stack
- IPv4, IPv6, Subnet Masking, Network Configuration and Troubleshooting
- Wi-Fi, Broadband, Email Usage
- Rules and Regulations in the Use of the Internet: Spam, Privacy, Intellectual Property Rights
- Email: Creating an Account, Sending and Accessing Messages, Attaching Documents
- Using the Internet: Social Networks, URLs, Search Engines, Downloading Information, Transferring Information to Other Applications
- Intellectual Property Rights, ICT Policy, Software Piracy

- Using Cloud Space: Google Drive, Google Workspace, OneDrive, Dropbox

Unit-8: Emerging Technologies — Data Analytics, Artificial Intelligence, Machine Learning [উদীয়মান প্রযুক্তি : তথ্য বিশ্লেষণ, কৃত্রিম বুদ্ধিমত্তা ও মেশিন লার্নিং]

- Introduction to Data Analytics: What is Data? Types of Data
- Basic Steps in Data Analytics, Simple Tools
- Artificial Intelligence (AI): Making machines capable of performing tasks requiring human-like thinking
- Common Examples: Voice Assistants, Facial Recognition, GPS Route Suggestions, Chatbots
- AI Capabilities: Understanding Language, Recognizing Patterns, Making Decisions
- Machine Learning (ML): Definition, How It Works, Classification and Examples
- Relationship Between Data Analytics, AI, and ML
- Benefits and Challenges of Data Analytics, AI, and ML
- Future Trends and Career Paths

Unit-9: Introduction to Freelancing [ফ্রিল্যান্সিং পরিচিতি]

- Introduction and Basic Concepts
- Fields of Freelancing
- Essential Skills and Preparation
- Introduction to Freelancing Market
- Building a Freelancing Career
- Challenges, Problems, and Solutions

List of Books

- Lambert, L. & Frye, C. (2015). *Microsoft Office 2016 Step by Step*. Microsoft Press.
- Hassan, N. & Hijazi, R. (2017). *Digital Privacy and Security Using Windows: A Practical Guide*. Apress.
- Massan, S. A. (2024). *Data Analytics Made Accessible*.
- Russell, S. & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
- Alpaydin, E. (2020). *Introduction to Machine Learning* (4th ed.). MIT Press.

Course Details

- **Course Code:** 216002
- **Course Title:** Lab: Information and Communication Technology
- **Marks:** 25
- **Credits:** 1

Course Objectives

- To develop students' understanding and skills in using technology to solve problems and enhance aspects of life and work.
- To prepare PowerPoint slides.
- To produce Word and Excel documents.

Economics Extracted Syllabus Content

Topic 1. Fundamentals of Economics

- **Contents:** Definition, Nature and Scope of Economics; Scarcity of Resources; Various forms of Economics; Positive and Normative Economics; Problems of Choice; Efficiency and Equity; Informal Economics.
- **Teaching Strategy:** Lecture and Assignment
- **CLOs:** CLO1

Topic 2. Supply and Demand

- **Contents:** Demand and Quantity Demanded, Determinants of Quantity Demanded, Demand Curve, Demand Schedule, Supply and Quantity Supplied, Determinants of Supply, Supply Schedule, Supply Curve, Movement along the Supply and Demand Curve and Shift of Supply and Demand Curve and Its Effects on Equilibrium Price and Quantity; Elasticity of Demand; Coefficient of Elasticity; Cross Elasticity.
- **Teaching Strategy:** Lecture, Group Discussion, Problem Solving and Assignment
- **CLOs:** CLO2, CLO5

Topic 3. Theory of Consumer Behavior

- **Contents:** Cardinal and Ordinal Utility Analyses; Total and Marginal Utility; Law of Diminishing Marginal Utility; Equi-Marginal Utility; Consumer Surplus.
- **Teaching Strategy:** Lecture, Problem Solving, Tutorial and Assignment
- **CLOs:** CLO3

Topic 4. Production and Cost

- **Contents:** Production Function and Technology; Production with One Variable Input; Production with Two Variable Inputs; Returns to Scale; Costs in the Short-run; Costs in the Long-run; LAC, SAC, LMC, SMC.
- **Teaching Strategy:** Lecture, Group Discussion and Tutorial
- **CLOs:** CLO4

Topic 5. Market Analysis

- **A. Perfect Competition:** Definition of Market; Characteristics; Average and Marginal Revenue; Individual and Market Supply; Short-run equilibrium of a competitive firm; Long-run Equilibrium; Causes of Disequilibrium.
- **B. Monopoly:** Characteristics; Average and Marginal Revenue; Supply Curve; Equilibrium Position; Comparison with Perfect Competition.
- **Teaching Strategy:** Lecture, Problem Solving, Group Discussion and Assignment
- **CLOs:** CLO2, CLO5

Topic 6. Overview of Macroeconomics

- **Contents:** Objectives and Instruments; National Income Accounting Methods; GDP; Double Counting Problem; NDP; GNP; From GDP to Disposable Income; Net Economic Welfare (NEW).
- **Teaching Strategy:** Lecture, Practical Example and Assignment
- **CLOs:** CLO6

Topic 7. International Trade

- **Contents:** Domestic vs. International Trade; Balance of Trade vs. Balance of Payment; Trends in Bangladesh; Free Trade vs. Protection; Absolute Advantage Theory; Comparative Advantage Theory.
- **Teaching Strategy:** Lecture, Group Discussion and Assignment
- **CLOs:** CLO7

Topic 8. Growth and Development

- **Contents:** Economic Development vs. Economic Growth; Measurement; Obstacles; Contemporary Concepts of Development.
- **Teaching Strategy:** Lecture, Group Discussion and Assignment
- **CLOs:** CLO8

Topic 9. Money

- **Contents:** Definition and Functions; Importance in Modern Economy; Concepts of Money (M_1 , M_2 , M_3); Value of Money; Inflation and Deflation (Concept, Causes, Effects).
- **Teaching Strategy:** Lecture, Group Discussion and Assignment
- **CLOs:** CLO9

Topic 10. Government Revenue and Expenditure

- **Contents:** Public vs. Private Finance; Sources of Govt. Revenues; Taxation (Direct & Indirect); Revenue Budget; Development Budget; Surplus, Deficit, Balanced Budget.
- **Teaching Strategy:** Lecture, Group Discussion, Practical Example and Assignment
- **CLOs:** CLO10

References (List of Books)

1. Case, K. E., & Fair, R. C. (2020). *Principles of Economics* (13th ed.). Pearson.
2. Mankiw, N. G. (2021). *Principles of Economics* (9th ed.). Cengage Learning